

Taehoon Hwang

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EDUCATION

Columbia University

B.S. in Computer Science; GPA: 4.05 / 4.33

New York, NY

Aug. 2025 – Present

Relevant Courses *Computer Science Theory, Programming Languages and Translators, Introduction to Databases, Projects in Computer Science, Theory ML Interaction, Natural Language Processing*

Purdue University

B.S. in Computer Science, Minor in Mathematics; GPA: 4.00 / 4.00

West Lafayette, IN

Aug. 2023 – May 2025

Relevant Courses *Linear Algebra, Probability, Discrete Mathematics, Computer Architecture, Data Structures, Statistical Methods, Competitive Programming I & II, Foundations of Deep Learning, Data Mining & Machine Learning, Introduction to Artificial Intelligence*

EXPERIENCE

Artificial Intelligence Researcher

Asteromorph

Jun. 2026 – Aug. 2026

Seoul, South Korea

- Developed and deployed a custom post-training engine for Evolutionary Strategies (ES) with an SGLang backend
- Matched the model performance of GRPO on verl, reducing end-to-end training time by 15%
- Supported major parallelism and quantization strategies across multi-node clusters, enabling post-training of models exceeding 1T parameters, including Kimi-K2.6, Kimi-K3 and Deepseek v4 Pro

Research Assistant

Columbia University

Sep. 2025 – Present

New York, NY

- Working on fine-grained advantage computations for reinforcement learning to improve GRPO methodologies

Artificial Intelligence Research Intern

Asteromorph

May 2025 – Aug. 2025

Seoul, South Korea

- Worked with fine-tuning and hosting LLMs (>200B parameters) on multi-GPU servers using vLLM and llama.cpp
- Developed an LLM fine-tuning framework by writing kernel code with PyTorch, Unsloth, and Hugging Face
- Deployed a parallel LLM fine-tuning scheduler and metric dashboard using Ray, Grafana, Prometheus, and Docker

Teaching Assistant

Purdue University

Jan. 2025 – May 2025

West Lafayette, IN

- Held office hours and lab sessions for students in Discrete Mathematics and Programming in C courses
- Reviewed and conducted problem solving sessions for discrete mathematics on first order logic, set theory, etc
- Answered students' questions during lab sessions on UNIX systems, file IO, dynamic memory allocation, etc

Research Assistant

Purdue University

Jan. 2024 – Dec. 2024

West Lafayette, IN

- Engineered demand forecasting pipeline integrating probability curves with LSTM encoding for enhanced accuracy
- Optimized inference storage by achieving a reduction of >99% compared to traditional deep learning models
- Conducted latent space analysis and illustrated composite encoding of demand features using LSTM models

Machine Learning Intern

Quantum Research Sciences

Sep. 2023 – Aug. 2024

West Lafayette, IN

- Employed ML algorithms and statistical techniques to deploy software for the United States Air Force
- Produced Monte Carlo simulations to test and tune quantum algorithms for inventory management
- Developed end-user interface for data analytics in quantum inventory management software

Research Assistant

Sungkyunkwan University

May 2021 – Dec. 2021

Seoul, South Korea

- Engaged with research team to boost open set recognition in ResNet models by 9%p via latent space manipulation
- Developed and deployed a remote automated testing and optimization framework for models using PyTorch
- Presented research findings at the Korean Science High School R&E Conference

PUBLICATIONS & CONTRIBUTIONS

LiveNewsBench: Evaluating LLM Web Search Capabilities <i>Yunfan Zhang, Kathleen McKeown, Smaranda Muresan (Contributor)</i>	May. 2026 <i>ICML 2026</i>
Spacer: Towards Engineered Scientific Inspiration <i>Minhyeong Lee et al. (Co-author)</i>	Aug. 2025 <i>arXiv Preprint</i>

PROJECTS

better-codex • Reengineered the Codex CLI as a full-screen Rust TUI with a dashboard and session-focused workspace • Built thread and session management for searching, resuming, forking, archiving, and deleting agent work • Integrated live tool output, command approvals, sandbox controls, and git diff review into a unified interface	Jun. 2026 – Present
Graduate School Research Agent • Developed agentic system to research graduate school programs, including fitness, quality, and difficulty • Implemented agentic retrieval loop, LLM quality judging, CV-aware fit assessment, and gap-fill passes using the Anthropic SDK • Added rich reporting using markdown files, full trajectory viewing and analysis, and interactive UIs	Apr. 2026 – Present
Stack Rotation • Developed full-rank fine-tuning LoRA methodology with PyTorch using stochastically sampled rotation matrices • Achieved SOTA-comparable performance when compared to LoRA variants such as DoRA, VeRA, and HiRA • Constructed parallel fine-tuning orchestrator using Ray, Grafana, Prometheus, and Docker	May 2025 – Aug. 2025
Rotationally Equivariant Spatio-temporal Prediction • Developed spatio-temporal predictive models with rotational equivariance to adapt to out-of-distribution inputs • Constructed training infrastructure using PyTorch Lightning and distributed high-performance computing nodes • Compared methods for equivariance using steerable wavelet filters, G-CNNs, and equivariant attention modules	Feb. 2025 – Apr. 2025
Financial Anomaly Detection with Modified Benford’s Law • Suggested novel alteration of Benford’s Law applicable to stock return data based on the Student-Lévy process • Proved algorithm utility with real-life stock data and demonstrated 60% improvement over conventional methods • Utilized new algorithm to detect financial anomalies in stock returns based on fitted location-scale t-distributions	Jan. 2025 – Mar. 2025
WUMT: Wavelet U-Net Motion Transformer • Investigated spatio-temporal encoding methods of videos using discrete wavelet transforms and NAFNet blocks • Utilized 4D motion tensor computations and predictions with transformers for U-Net latent space operations • Built training infrastructure with PyTorch Lightning and leveraged Docker and MLFlow for streamlined research	Jul. 2024 – Mar. 2025
Contextual-Diffusion • Improved cohesion and features of Stable Diffusion model output utilizing spacial context from LLMs • Developed pipeline incorporating Mask-RCNN models as translation layers between LLMs and Stable Diffusion • Generated and annotated image segmentation datasets for supervised learning of Mask-RCNN models	Feb. 2023 – Aug. 2023
Minimax-based Animal Shogi AI • Programmed an Animal Shogi bot using a minimax algorithm with alpha-beta pruning in C++ • Implemented interactable game GUI and cross-language translation mechanism with Python • Demonstrated bot with live play-testing at school festival to 350+ students and faculty	Mar. 2021 – Nov. 2021

ACTIVITIES

Purdue Hackers Club Member

Jan. 2025 – May 2025

Purdue University

West Lafayette, IN

- Participated in weekly 'hack' sessions and developed projects involving full-stack web development.
- Developed website for course scheduling at Purdue with Next.js, MongoDB, and Python among team of 3.

Purdue IGDC Club Member

Jan. 2024 – May 2025

Purdue University

West Lafayette, IN

- Participated in various in-person game development sessions with other club members.
- Developed game with Unity centered around creating gravitational wells to induce fast-paced gameplay.
- Gave feedback on other games being developed and participated in play testing.

ICPC 2023 ECNA Regional Participant

Sep. 2023

Purdue University

West Lafayette, IN

- Ranked in the Top 20 in Purdue's ICPC participant selection contest
- Lead team of 3 in the ICPC East Central North America regional contest among 5 other Purdue ICPC teams
- Solved problems utilizing competitive programming techniques such as dynamic programming and backtracking

Hello World Hackathon Participant

Sep. 2023

Purdue University

West Lafayette, IN

- Led a team of 4 in a 24-hour hackathon driving project development and collaboration
- Built a software pipeline utilizing LLMs to deliver personalized dietary text feedback to users
- Built a full-stack web app with a React front-end and ExpressJS back-end integrating MongoDB RestAPIs

ML@Purdue Club Member

Mar. 2023 – May 2025

Purdue University

West Lafayette, IN

- Engaged in club discourse on AI and ML topics for research and software development.
- Participated in paper reading and workshops on various topics, such as transformers, GNNs, etc.

Published Author on Neural Network Fundamentals

Sep. 2021 – Mar. 2022

Barun Books Co.,Ltd

Seoul, South Korea

- Authored a 160-page book on neural networks covering backpropagation, gradient descent, and parallelization
- Distributed to 13 retailers, selling 300+ copies in the first year
- Ranked as a top 4 entry in the "Weekly Top Releases" by the second-largest South Korean book retailer

HONORS & AWARDS

Dean's List

Columbia University, 2025 & 2026

Purdue Office of Undergraduate Research Grant

Purdue University, 2024

\$ 500

Dean's List & Semester Honors

Purdue University, 2023 & 2024 & 2025

Excellence Award in the Gifted School Research Conference

Pohang University of Science & Technology, 2022

Azure AI Fundamentals Certification

Microsoft, 2022

TECHNICAL SKILLS

Languages & Frameworks: C/C++, Express.js, Flask, HTML/CSS, Java, JavaScript, Python, React, x86 Assembly

Dev & Research Tools: Docker, Git/GitHub, IntelliJ, LaTeX, MLfLow, Overleaf, PyCharm, Slurm, VS Code

Libraries: Keras, NumPy, OpenCV, Pandas, PyTorch, Scikit-learn, SGLang, Transformers, Unsloth, trl, vLLM